

EXTREMAL GRAPH THEORY

Erdős Problem 915

Maximizing Edge Density Subject to Connectivity Constraints

One-hour version

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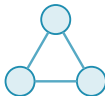
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KU Leuven, Master of Mathematics

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The eight models



simple graph

one edge per pair



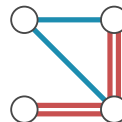
multigraph

parallel edges $\mu(u, v) \geq 1$



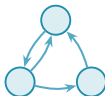
hypergraph

hyperedges span $r \geq 3$



multi-hypergraph

a hyperedge repeated



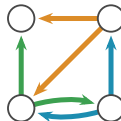
simple digraph

one arc per ordered pair



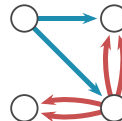
directed multigraph

parallel and opposite arcs



directed hypergraph

two share an edge

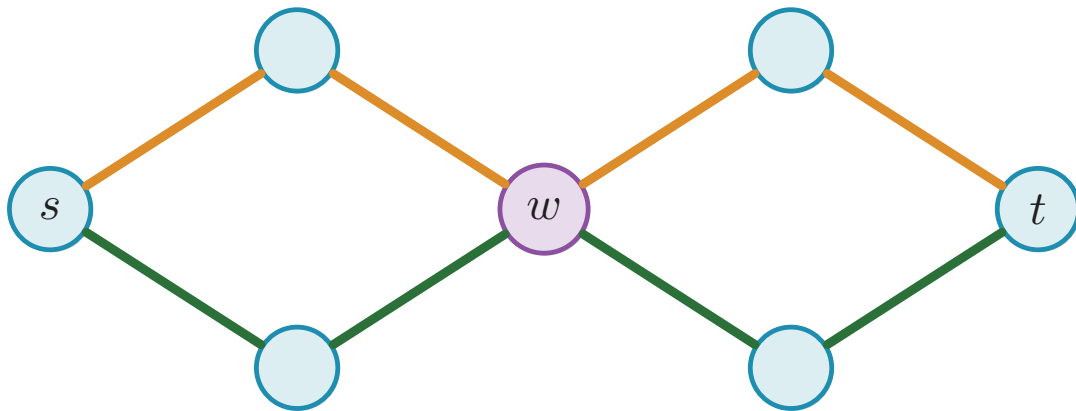


directed multi-hypergraph

a hyperedge repeated

The eight models side by side, undirected on top and directed below. The simple graph is a triangle, the multigraph has parallel edges, the hypergraph has three hyperedges, and the multi-hypergraph has one repeated hyperedge. The directed versions have arcs instead of edges, with the same pattern of repetition and hyperedges.

The separation axis

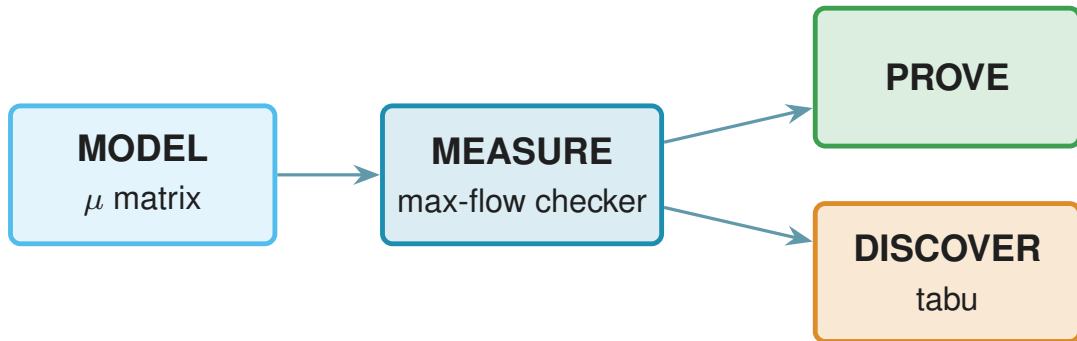


Two edge-disjoint s – t routes both pass through w , so $\lambda(s, t) = 2$ but $\kappa(s, t) = 1$.

Notation for the extremal functions

Model	edge-disjoint routes	internally vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r -uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r -uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\text{dir}}(n)$	$k_m^{\text{dir}}(n)$
Directed multigraph	$L_m^{\text{dir}}(n)$	$K_m^{\text{dir}}(n)$
Directed r -uniform simple hypergraph	$\ell_m^{\text{dir}(r)}(n)$	$k_m^{\text{dir}(r)}(n)$
Directed r -uniform multihypergraph	$L_m^{\text{dir}(r)}(n)$	$K_m^{\text{dir}(r)}(n)$

The architecture

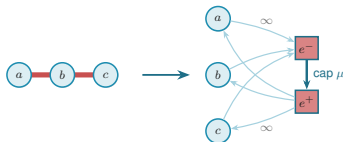


A single model feeds the exact max-flow checker. Certification turns that measurement into finite upper bounds, while heuristic search uses it to discover lower bounds.

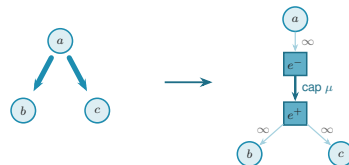
The building blocks of the connectivity checker



Split each vertex v into v^{in} and v^{out} joined by a unit-capacity arc to transform vertex-disjointness into edge-disjointness.

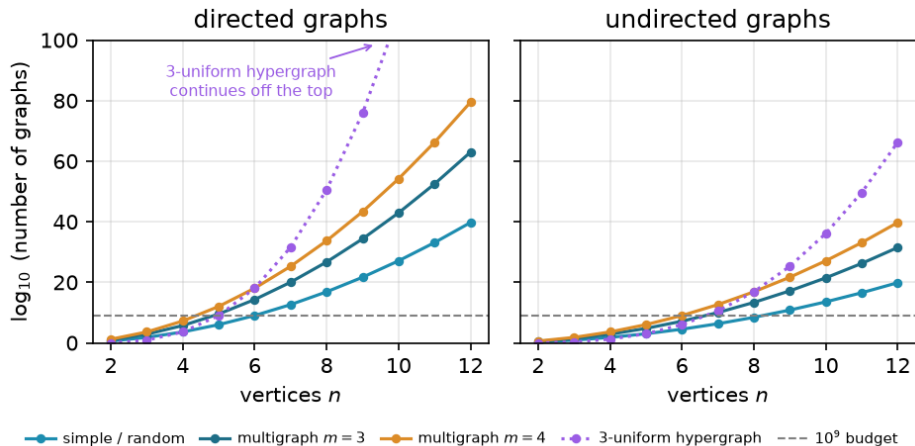


Transform a hyperedge into a gate with capacity equal to the multiplicity of the hyperedge.



The directed hyperedge becomes a gate with one tail and two heads; capacity equal to the multiplicity, and every route must enter at the tail and leave at a head.

Candidate counts

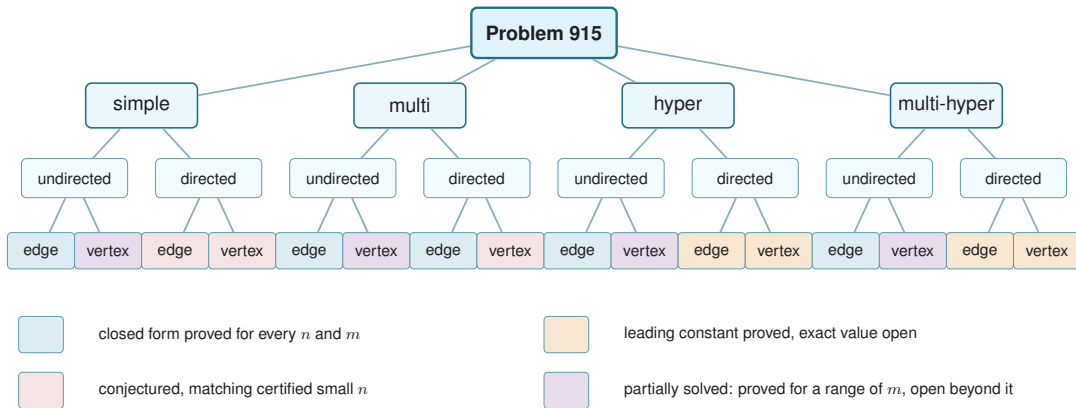


Candidate counts for blind enumeration on a logarithmic scale. Every model crosses the generous 10^9 budget in single-digit order, and directed objects grow fastest because they have more independent entries to decide.

Generating search: a benchmark

n	$\ell_2^{\text{dir}}(n)$	blind (s)	pruned (s)	generated (s)
3	4	0.05	0.003	0.009
4	6	6.3	0.008	0.008
5	8	—	0.22	0.039
6	10	—	14.5	0.73
7	12	—	1156.6	17.0

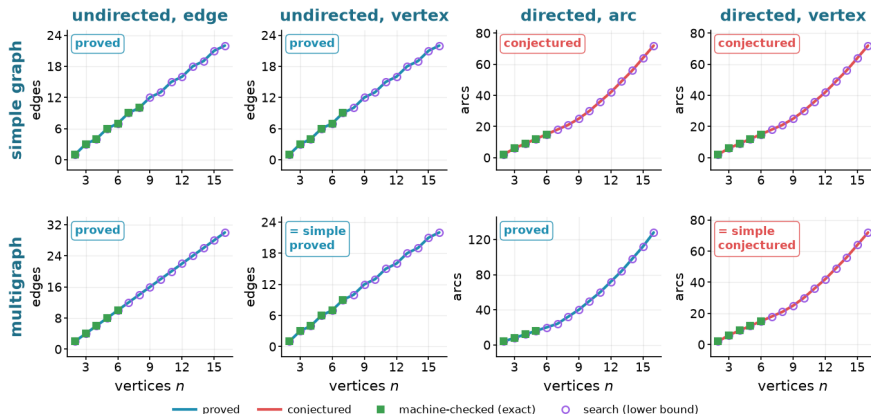
Status of the sixteen variants



Status of the sixteen variants. Colour marks the status of a leaf: blue a closed form proved for every n and m , purple a value proved for a range of m and open beyond it, red a conjectured exact formula with a proved lower bound, and amber a proved leading term with the exact value unknown.

Eight graph variants at $m = 3$

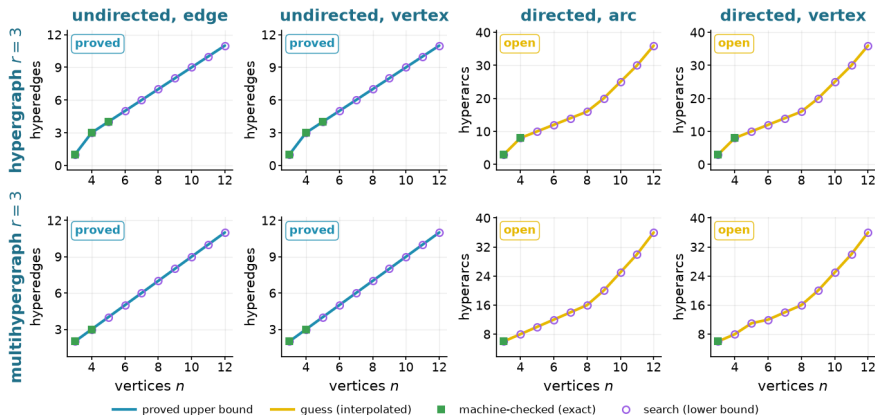
Erdős 915 across the sixteen variants, $m = 3$ · simple and multigraph models



The eight graph-model variants at $m = 3$: simple and multigraph rows, undirected and directed columns. A blue curve is a theorem, a red one a conjecture and a yellow one an interpolation, filled squares are exact finite values, and open circles are witnesses.

Eight hypergraph variants at $m = 3$

Erdős 915 across the sixteen variants, $m = 3$ · hypergraph and multihypergraph models

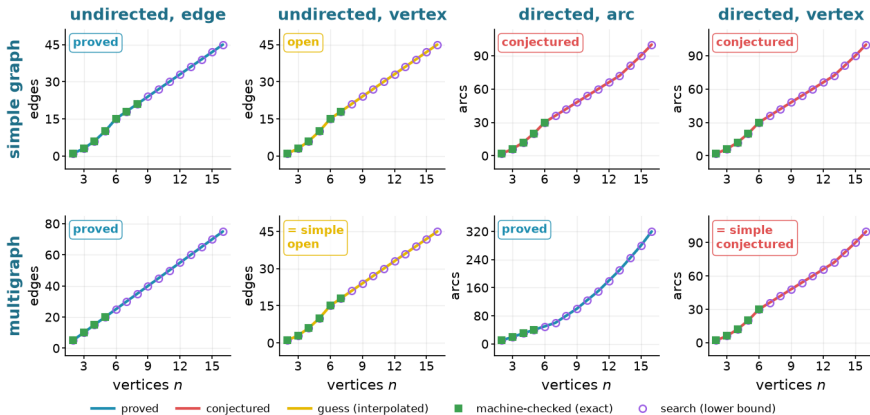


The eight hypergraph-model variants at $m = 3$: hypergraph and multihypergraph rows, undirected and directed columns, same colour and marker key.

The blue curve is a proved upper bound rather than a proved value, attained only inside the stated ranges, so a square or circle below it marks a gap that is open rather than an error.

Eight graph variants at $m = 6$

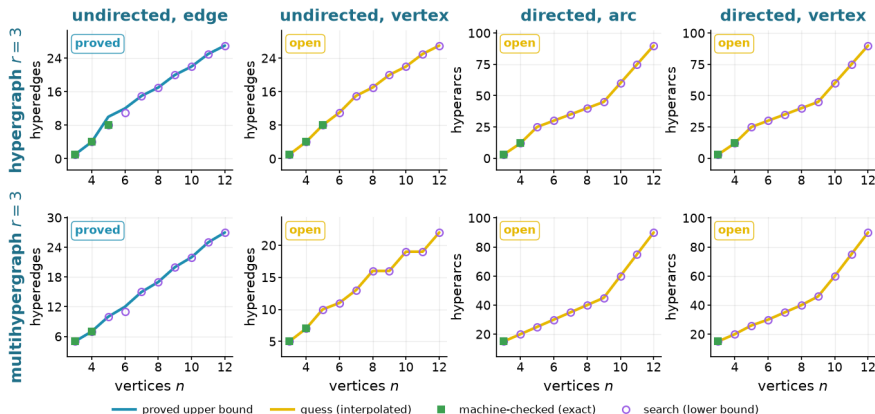
Erdős 915 across the sixteen variants, $m = 6$ · simple and multigraph models



The same eight graph-model variants at $m = 6$. Raising m widens every feasible band, and the exact squares thin out as the enumeration range falls behind the constructions.

Eight hypergraph variants at $m = 6$

Erdős 915 across the sixteen variants, $m = 6$ · hypergraph and multihypergraph models



The same eight hypergraph-model variants at $m = 6$. The blue curve is again the proved upper bound, and at this m it runs visibly above what has been reached: the exact square at $n = 5$ is 8 against the curve's 10, and at $n = 6$ the curve reads 12 where the true maximum is 11.

The sixteen variants, summarised

Model	Edge or arc-disjoint	Vertex-disjoint
<i>Undirected</i>		
simple graph	$\lfloor m(n-1)/2 \rfloor$, proved	the edge value for $m \leq 4$, and $\lfloor 8n/3 \rfloor - 4$ at $m = 5$; open for $m \geq 6$
multigraph	$(m-1)(n-1)$, proved	the simple value
r -hypergraph	$\leq \lfloor (m-1)(n-1)/(r-1) \rfloor$, with equality in the range below	$\lfloor (n-1)/(r-1) \rfloor$ at $m = 2$ and $\leq \lfloor 2(n-1)/(r-1) \rfloor$ at $m = 3$; open for $m \geq 4$
r -multi-hyper	the same bound, with equality when $(r-1) \mid (n-1)$	the same bounds, repeats closing cases the simple model misses at $m = 3$
<i>Directed</i>		
simple digraph	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \geq 3$, exact value conjectured	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \geq 3$, equality with the arc value open
multigraph	$(m-1)M(n)$, proved	the simple digraph value
r -hypergraph	$(1+o(1))(m-1)n^2/[4(r-1)]$ for fixed r, m , leading term proved	the same leading term for fixed r, m
r -multi-hyper	the same leading term, the bound proved with repeats allowed	the same leading term for fixed r, m

Exact values, $m = 3$

Model	Variant	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	4	6	7	9	10
	undirected, vertex ($k_m(n)$), proved	1	3	4	6	7	9	10
	directed, arc ($\ell_m^{\text{dir}}(n)$), conjectured	2	6	9	12	15	≥ 18	≥ 21
	directed, vertex ($k_m^{\text{dir}}(n)$), conjectured	2	6	9	12	15	≥ 18	≥ 21
multigraph	undirected, edge ($L_m(n)$), proved	2	4	6	8	10	12	14
	undirected, vertex ($k_m(n)$), = simple, proved	1	3	4	6	7	9	10
	directed, arc ($L_m^{\text{dir}}(n)$), proved	4	8	12	16	20	24	32
	directed, vertex ($k_m^{\text{dir}}(n)$), = simple, conjectured	2	6	9	12	15	≥ 18	≥ 21
hypergraph $r = 3$	undirected, edge, proved		1	3	4	≤ 5	≤ 6	≤ 7
	undirected, vertex, proved		1	3	4	≤ 5	≤ 6	≤ 7
	directed, arc, open		3	8	≥ 10	≥ 12	≥ 14	≥ 16
	directed, vertex, open		3	8	≥ 10	≥ 12	≥ 14	≥ 16
multihypergraph $r = 3$	undirected, edge, proved		2	3	≤ 4	≤ 5	≤ 6	≤ 7
	undirected, vertex, proved		2	3	≤ 4	≤ 5	≤ 6	≤ 7
	directed, arc, open		6	≥ 8	≥ 10	≥ 12	≥ 14	≥ 16
	directed, vertex, open		6	≥ 8	≥ 11	≥ 12	≥ 14	≥ 16

Exact values, $m = 6$

Model	Variant	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	6	10	15	18	21
	undirected, vertex ($k_m(n)$), open	1	3	6	10	15	18	≥ 21
	directed, arc ($\ell_m^{\text{dir}}(n)$), conjectured	2	6	12	20	30	≥ 36	≥ 42
	directed, vertex ($k_m^{\text{dir}}(n)$), conjectured	2	6	12	20	30	≥ 36	≥ 42
multigraph	undirected, edge ($L_m(n)$), proved	5	10	15	20	25	30	35
	undirected, vertex ($k_m(n)$), = simple, open	1	3	6	10	15	18	≥ 21
	directed, arc ($L_m^{\text{dir}}(n)$), proved	10	20	30	40	50	60	80
	directed, vertex ($k_m^{\text{dir}}(n)$), = simple, conjectured	2	6	12	20	30	≥ 36	≥ 42
hypergraph $r = 3$	undirected, edge, proved		1	4	8	≤ 12	≤ 15	≤ 17
	undirected, vertex, open		1	4	8	≥ 11	≥ 15	≥ 17
	directed, arc, open		3	12	≥ 25	≥ 30	≥ 35	≥ 40
	directed, vertex, open		3	12	≥ 25	≥ 30	≥ 35	≥ 40
multihypergraph $r = 3$	undirected, edge, proved		5	7	≤ 10	≤ 12	≤ 15	≤ 17
	undirected, vertex, open		5	7	≥ 10	≥ 11	≥ 13	≥ 16
	directed, arc, open		15	≥ 20	≥ 25	≥ 30	≥ 35	≥ 40
	directed, vertex, open		15	≥ 20	≥ 26	≥ 30	≥ 35	≥ 40

What remains open

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$	Prove the decomposition, or otherwise determine the linear coefficient.
Directed simple vertex, $m \geq 3$	$k_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$	Decide whether its exact value equals the arc value, given that an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$, where $c_m = \lim_n k_m(n)/(n-1)$	Determine c_m , whose likely next structure is triconnected rather than merely blockwise.
Multigraph vertex, parallel copies distinct	Block-knapsack formulation and order $\Theta(m^2 n)$ as $m \rightarrow \infty$ with $n/m \rightarrow \infty$, with the per-vertex constant between $3 - 2\sqrt{2}$ and 2	Determine the asymptotic constant and the optimal 2-connected blocks.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$	Locate the first failure for $r \geq 3$ and replace the rank-two decomposition used at $m = 3$.
Directed hypergraphs	For fixed r, m , leading term $(m-1)n^2/(4(r-1))$ for all orientations and both separations	Determine finite exact values and whether forward and general agree once $n > r$.

AI: curse or blessing?

Pros

- $\text{\LaTeX}$ typesetting
- length & complexity of generated text

Cons

- length & complexity of generated text
 - control
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It's a tool.

short leash + AI proofs = argued conjecture [AI]